

MTH603

Assignment 1

Solution by Vu360

Question No 1 Solution:

Let's find a root of the equation $x^3 - 3x - 5 = 0$ in the interval $(2, 3)$ using the Bisection Method after three iterations.

Initial interval: $(2, 3)$

Iteration 1:

$$\text{Midpoint: } x^1 = \frac{(2 + 3)}{2} = 2.5$$

$$f(x^1) = (2.5)^3 - 3(2.5) - 5 = -0.375$$

Since $f(x^1) < 0$, the root lies in the right half of the interval: $(2.5, 3)$

Iteration 2:

$$\text{Midpoint: } x^2 = \frac{(2.5 + 3)}{2} = 2.75$$

$$f(x^2) = (2.75)^3 - 3(2.75) - 5 = 0.359$$

Since $f(x^2) > 0$, the root lies in the left half of the interval: $(2.5, 2.75)$

Iteration 3:

$$\text{Midpoint: } x^3 = \frac{(2.5 + 2.75)}{2} = 2.625$$

$$f(x^3) = (2.625)^3 - 3(2.625) - 5 = -0.046$$

After three iterations, the approximate root is $x \approx 2.625$, accurate to four decimal places.

Note: The actual root is $x \approx 2.618$, but our approximation is close enough given the required accuracy.

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Question No 2 Solution:

Let's find a root of the equation $x e^x - \cos(x)$

= 0 in the interval (0,1) using the Newton

– Raphson Method after three iterations, starting with the initial value x^0

= 0.5.

$$\text{Newton – Raphson Formula: } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Iteration 1:

$$x^0 = 0.5$$

$$f(x^0) = (0.5)e^{0.5} - \cos(0.5) \approx 0.521$$

$$f'(x^0) = (0.5)e^{0.5} + \sin(0.5) \approx 1.283$$

$$x^1 = x^0 - \frac{f(x^0)}{f'(x^0)} \approx 0.5 - 0.521/1.283 \approx 0.438$$

Iteration 2:

$$x^1 = 0.438$$

$$f(x^1) = (0.438)e^{0.438} - \cos(0.438) \approx 0.034$$

$$f'(x^1) = (0.438)e^{0.438} + \sin(0.438) \approx 1.146$$

$$x^2 = x^1 - \frac{f(x^1)}{f'(x^1)} \approx 0.438 - 0.034/1.146 \approx 0.414$$

Iteration 3:

$$x^2 = 0.414$$

$$f(x^2) = (0.414)e^{0.414} - \cos(0.414) \approx 0.002$$

$$f'(x^2) = (0.414)e^{0.414} + \sin(0.414) \approx 1.093$$

$$x^3 = x^2 - \frac{f(x^2)}{f'(x^2)} \approx 0.414 - 0.002/1.093 \approx 0.413$$

After three iterations, the approximate root is x

≈ 0.413 , accurate to four decimal places.

Note: The actual root is x

≈ 0.41279 , but our approximation is close enough given the required accuracy.